

PERCEIVED IMPACT OF WEB-BASED LEARNING ENVIRONMENTS ON ACADEMIC ACHIEVEMENT OF BUSINESS EDUCATION UNDERGRADUATES

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Abstract

The study examined the perceived impact of Web-based learning environments on academic achievement of business education undergraduates. Three research questions were answered. A descriptive survey research design was used. The population of the study comprised 1,926 Business Education undergraduate students of Tai Solarin University of Education (TASUED) for the academic session 2019/2020. A total of 500 business education undergraduates were selected as study sample. Simple random sampling procedure was used for the selection of the sample. Researchers' Instrument tagged 'Web-Based Learning Environments and Academic Achievement Questionnaire (WBLEAAQ)' was used for data collection. The reliability of the instrument was conducted among 20 business education undergraduates twice in Olabisi Onabanjo University (OOU), Ogun State and the data generated was subjected to Pearson Product Moment Correlation (PPMC) formula and which yielded 0.79 as reliability coefficient. Business education undergraduates CGPA were collected to measure their academic achievement. Descriptive statistics of mean was used for answering research questions one and two. Regression analysis was used for answering research question 3. The study findings shown that ability to access information in different area, efficient way to deliver course materials, access to resources from any location, increase the potential for widening access, promote social development, enhance interactive learning based and effective in time management were among the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement. Inadequate undergraduates' motivation and engagement, poor state of infrastructure, inadequate financial resources, lack of learner motivation, students isolated problems, inequality in access to the technology by undergraduates, poor nature of maintenance culture and electricity supply issue were among the challenges confronting Web-based learning in public university environments. It was also revealed that the application of Web-based learning for instructional delivery could result to an increase in academic achievement of business education undergraduates. It was recommended that government and other stakeholders in university education should provide adequate financial resources for the purchase and installation of needed Web-based learning in system. Adequate training should be provided for the lecturers to handle Web-based learning very effective.

Keywords: Web-Based Learning, Academic Achievement, Business Education, Undergraduates

Introduction

The ultimate goal of educational system is to see students excel in their academic endeavour. This form part of the reasons government, parents, and other stakeholders in educational industry are spending huge amounts of resources on the development of the sector. Academic achievement is the average academic outcomes of students in a given semester or session. Undergraduates' academic achievement in Nigeria public universities seems to be falling as reflected in the quality of graduates produced every year. Opinions poll from different quarters have suggested that undergraduates' academic achievement are being represented by the classes of degrees graduated at in which business education students' were not an exception.

Business education is the educational programme used in training undergraduates on saleable skills to provide solutions to office work and economic self-reliance (Okoro, 2013). It is also an educational programme aimed to prepare students employability skill for gainful employment opportunity in both the private and public organizations (Nwaigburu & Eneogwe, 2013).

Web-based learning is among digital technology tools that when used effective, it could engender learning outcomes in form of improved academic achievement. It provides the students' easy access to information at convenient time, enhances internet data sharing, encourage interactive academic activities, and allow distance education learning (Nordin 1. Alias, 2019). Shih-Wei and Chien-Hung (2017) said that Web-based learning when used appropriately might have dramatic effect on mode of instructional delivery. Web-based learning provides undergraduates' access to instructional information, than would lecture mode of instruction. It enhances learning experiences that are welcome by the learners for its openness, ease teaching process and made learning more clearly. It also serves as avenue to generate opportunities for engagement and guarantee efficient instructional delivery. Web-based learning has the capacity to achieve learning aims and objectives of the learners. As stated by Anunobi, Gambari, Abdullahi and Alabi (2016), the effectiveness of Web-based learning to a larger extent depends on networking operation systems. These might also enhance enrolment into courses, course/program of instruction, carrying out assignments, discussing with instructors and fellow undergraduates.

Egoigwe *et al* (2020) reiterated that electronic instruction delivery has direct effect on postgraduate students' academic performance in university and that inadequate ICT tools, inadequate qualified teaching staff and skills to teach ICT, poor nature of internet connectivity, fear of redundant by teaching staff were among the problems confronting the use of electronic learning on postgraduate students in Nigeria. Nwazor and Godwin-Maduikwe (2015) asserted that business education students' usages of social media were not for academic purpose while Anunobi *et al* (2016) stated that students taught with Web-based instruction performed and retained academic concepts than that of counterpart taught with lecture instructional method. Eze, Chinedu-Eze and Bello (2018) found that electronic learning facilities e-learning promotes interactive academic activities and enhance ease

access to information. However, they also revealed that lopsided attitudes of teachers, poor nature of internet availability, poor training of teachers were among the problems facing the effective utilization of e-learning towards improving students' academic performance. Basri, Alandejani and Almadani (2018) concluded that adoption of ICT tools and university students' academic performance directly related in Saudi Arabia.

Statement of the Problem

Despite the benefits of digital technology tools application on academic achievement of university students, records seem to show that poor academic achievement persists as reflected in the education quality produces by the university nowadays. Many arguments have been provided on the factors that may have caused this low level of achievement. Some believed undergraduate study habit, lack of parental care, un-conducive teaching environment, inadequate teaching aid among others are responsible factors that may have caused this. A common observation of the university education system in Nigeria would show a poor state of infrastructure needed for effective instructional delivery. It is thus not too clear the extent to which such variables such as Web-based learning environments can account for improved academic achievement of business education undergraduates. Hence, the need for this study.

Purpose of the Study

The main purpose of the study was to examine the perceived impact of Web-based learning environments on academic achievement of business education undergraduates. Specifically, the study:

1. Identified the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement.
2. Investigated the challenges confronting Web-based learning in public university environments.
3. Examined the extent Web-based learning can impact academic achievement of business education undergraduates.

Research Questions

1. What are the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement?
2. What are the challenges confronting Web-based learning in public university environments?
3. To what extent can Web-based learning impact academic achievement of business education undergraduates?

Methodology

The study adopted a descriptive survey research design. This study was conducted in Tai Solarin University of Education (TASUED), Ijagun, Ogun State. The population of the study comprised 1,926 Business Education undergraduate students for the academic

session 2019/2020. A total of 526 undergraduates were from 100 level, 560 from 200 level, 570 from 300 level and 270 from 400 level. A total of 500 business education undergraduates were selected as study sample size. Simple random sampling procedure was used for the selection of the sample. Researchers' Instrument tagged 'Web-Based Learning Environments and Academic Achievement Questionnaire (WBLEAAQ)' was used for data collection. The instrument was validated by three experts from Measurement and Evaluation Departments, TASUED. The reliability of the instrument was conducted among 20 business education undergraduates twice in Olabisi Onabanjo University (OOU), Ogun State and the data generated was subjected to Pearson Product Moment Correlation (PPMC) formula and which yielded 0.79 as reliability coefficient. Business education undergraduates CGPA were collected to measure their academic achievement. Descriptive statistics of mean were used for answering research questions one and two. Regression analysis was used for answering research question 3. **Decision Criteria:** Based on the cut-off point of 2.50 (strongly agree (SA) 4, agree (A) 3, disagree (D) 2 and strongly disagree (SD) 1). Any mean score equal to 2.50 or greater than 2.50 was regarded as agreed while any items with less than 2.50 was disagreed.

Results and Discussion

Research Question 1: What are the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement?

Table 1: Mean responses on the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement

S/N	Items	Mean	Remarks
1.	Ability to access information in different area.	3.196	Agreed
2.	Efficient way to deliver course materials.	2.700	Agreed
3.	Access to resources from any location.	3.000	Agreed
4.	Increase the potential for widening access.	2.952	Agreed
5.	Encourage social development	2.986	Agreed
6.	Encourage interactive learning based	3.190	Agreed
7.	Effective in time management	2.812	Agreed

Source: Field Survey, 2021

Table 1 reveal that ability to access information in different area, efficient way to deliver course materials, access to resources from any location, increase the potential for widening access, encourage social development, interactive learning based and effective in time management were among the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement.

Research Question 2: What are the challenges confronting Web-based learning in public university environments?

Table 2: Mean responses on the challenges confronting Web-based learning in public university environments

S/N	Items	Mean	Remarks
1.	Inadequate undergraduates' motivation and engagement	3.132	Agreed
2.	Poor nature of infrastructure	3.069	Agreed
3.	Inadequate financial resources	3.136	Agreed
4.	Lack of learner motivation	2.983	Agreed
5.	Students isolated problems	2.660	Agreed
6.	Inequality access to the technology by undergraduates	2.992	Agreed
7.	Poor nature of maintenance culture	3.210	Agreed
8.	Electricity supply issue	2.559	Agreed

Source: Field Survey, 2021

Table 2 reveal that inadequate undergraduate's motivation and engagement, poor nature of infrastructure, inadequate financial resources, lack of learner motivation, students isolated problems, inequality access to the technology by undergraduates, poor nature of maintenance culture and electricity supply issue were among the challenges confronting Web-based learning in public university environments.

Research Question 3: To what extent can Web-based learning impact academic achievement of business education undergraduates?

Table 3: Extent Web-based learning impact academic achievement of business education undergraduates

Model	Unstandardized Coefficients		Standardized Coefficients	T	Sig.
	B	Std. Error	Beta		
1 (Constant)	5.123	3.005		1.705	.089
Web-based learning	.350	.089	.222	3.931	.000

a. Dependent Variable: Academic achievement of business education undergraduates Table reveal that the sign of the coefficient of Web-based learning is positive. This implied that the application of Web-based learning for instructional delivery could result to an increase in academic achievement of business education undergraduates. The probability ($p = 0.00$) as reported in Table 3 for Web-based learning implied that the slope ($= 0.222$) is statistically significant. Hence, the researcher concluded that about 22.2% of an increase in academic achievement of business education undergraduates could be attributed towards application of Web-based learning.

Discussion of Findings

This study findings shown that ability to access information in different area, efficient way to deliver course materials, access to resources from any location, increase the potential for widening access, encourage social development, interactive learning

based and effective in time management were among the perceptions of business education undergraduates on the benefits of Web-based learning environments on academic achievement. These findings corroborate with Eze, Chinedu-Eze and Bello (2018) found that electronic learning facilities promotes interactive academic activities and enhance ease access to information. Mothibi (2015) found that the effectiveness electronic learning and students' performance are directly related. Chou and Liu (2017) findings indicated that students taught with Web-based learning developed more computer self-efficacy and academic achievement.

The study findings indicated that inadequate undergraduate's motivation and engagement, poor nature of infrastructure, inadequate financial resources, lack of learner motivation, students isolated problems, inequality access to the technology by undergraduates, poor nature of maintenance culture and electricity supply issue were among the challenges confronting Web-based learning in public university environments. These findings correlate with Egoigwe et al. (2020) revealed that inadequate ICT tools, inadequate qualified teaching staff and skills to teach ICT, poor nature of internet connectivity, fear of redundant by teaching staff were among the problems confronting the use of electronic learning on postgraduate students in Nigeria. Eze, Chinedu-Eze and Bello (2018) found that lopsided attitudes of teachers, poor nature of internet availability, poor training of teachers were among the problems facing the effective utilization of e-learning towards improving students' academic performance.

It was also revealed that the application of Web-based learning for instructional delivery could result to an increase in academic achievement of business education undergraduates. These findings corroborate with Egoigwe et al (2020) who indicated that electronic instruction delivery has direct effect on postgraduate students' academic performance in university. Anunobi et al (2016) indicated that students taught with Web-based instruction performed and retained academic concepts than that of counterpart taught with lecture instructional method.

Conclusion

The adoption of ICT tools such as Web-based learning could predict an improvement in business education undergraduates. However, this study concluded based on the findings that with the provision of adequate infrastructure and ICT tools needed for the effective operation of Web-based learning, about 22.2% of an increase in academic achievement of business education undergraduates could be attributed towards application of Web-based learning.

Recommendations

Based on these findings, the following recommendations are provided:

- a. Government and other stakeholders in university education should provide adequate financial resources for the purchase and installation of needed Web-based learning in system.

- a. Adequate training should be provided for the lecturers to handle Web-based learning very effective.
- b. Provision of stable electricity in the country should be ensuring by the government.
- c. Government through the university authority should approve the monthly financial allowance for undergraduates and lecturers.
- d. The university authority should organize training and workshop for undergraduates to showcase the benefits of Web-based learning and motivation for them.

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